MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE
EXAMINATION APRIL 2026
SECOND SEMESTER
Discipline Specific Core Course (DSC) - MG2DSCICH100 - MAJOR
CHEMICAL INDUSTRIES
(2024 ADMISSION ONWARDS)
Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A
Multiple choice questions.Answer all questions
Each question carries 1 marks

1.What is the primary cause of resistance in normal conductors, which disappears in superconducting oxides? [U] / [CO1]
d). Lattice vibrations

2.Which of the following types of glass typically contains metal oxides or sulphides to impart color? [K] / [CO2]
c). Coloured glass

3.In solar cell, which energy is getting converted into electricity. [K] / [CO2]
b). light energy

4.Why is the Nernst equation important in biological systems? [U] / [CO3]
c).It determines membrane potentials

5.The overall reaction in a standard Daniel cell is: [An] / [CO3]
a). $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

6.Armoured glass achieves its strength through [A] / [CO1]
b). The use of multiple glass- coatings polymer layers

7.What is the role of the die in extrusion moulding? [U] / [CO4]
b). Shapes the plastic into its final form

8.Which fraction is obtained first during the distillation of coal tar? [U] / [CO1, CO5]
b). Light oils

9.What is the primary goal of the Fischer-Tropsch process? [U] / [CO5]
b). To convert carbon monoxide and hydrogen into hydrocarbons

10.What is the primary purpose of the injection screw in injection moulding? [U] / [CO4]
c). To melt and feed the plastic material into the mould

Part B
Short Answer Questions.Answer any 4 questions
Each question carries 3 marks

11.Describe the formulation process of water-based and solvent-based paints. [A] / [CO1]
The formulation of paints involves combining pigments, binders, solvents or water, and various additives to obtain the desired properties such as color, adhesion, durability, and drying time. In water-based paints, water acts as the main dispersion medium. A water-dispersible binder, such as an acrylic or vinyl emulsion, is then added to form the film on drying. Various additives like thickeners, defoamers, preservatives, and coalescing agents are incorporated to improve stability, application properties, and shelf life. In solvent-based paints, organic solvents such as mineral spirits are used as the medium. The pigments are first ground with a portion of the binder, typically an alkyd resin, to achieve fine dispersion. The remaining binder and solvent are then added to adjust consistency and improve flow. Additives such as driers, plasticizers,

and anti-skinning agents are included to enhance drying and performance. (Any three points – 3 marks)

12.What is the main chemical component of silicate glass?[U] / [CO2]
Silicate glass is primarily composed of silicon dioxide (SiO_2)**,** which forms the basic network structure of the glass. In addition to silica, other components are added to modify its properties. These include alkali metal oxides such as sodium oxide (Na_2O) or potassium oxide (K_2O), which act as fluxes and lower the melting temperature of silica, making the glass easier to shape. Alkaline earth oxides, particularly calcium oxide (CaO), are added as stabilizers to improve the chemical durability and mechanical strength of the glass. Sometimes, small amounts of other oxides like aluminum oxide (Al_2O_3), magnesium oxide (MgO), or boron oxide (B_2O_3) are included to further enhance properties such as hardness.
(Any three components – 3 marks)

13.What are the main components of a battery? [U] / [CO3]
The main components of a battery include the following:
Anode (negative electrode): This is the electrode where oxidation occurs during discharge. It releases electrons to the external circuit.
Cathode (positive electrode): This is the electrode where reduction occurs. It accepts electrons from the external circuit.
Electrolyte: A chemical medium (liquid, solid, or gel) that allows the movement of ions between the anode and cathode, enabling the electrochemical reaction.
Separator: A porous insulating material placed between the anode and cathode to prevent direct contact (short circuit) while allowing ion flow.
Current collectors: Conductive materials (usually metals) that collect and transfer electrons between the electrodes and the external circuit.
(Components with explanation -3 marks)

14.What are the main components of a battery and their functions? [K] / [CO3]
The main components of a battery include the following:
Anode (negative electrode): This is the electrode where oxidation occurs during discharge. It releases electrons to the external circuit.
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Separator: A porous insulating material placed between the anode and cathode to prevent direct contact (short circuit) while allowing ion flow.
Current collectors: Conductive materials (usually metals) that collect and transfer electrons between the electrodes and the external circuit.
(Any three components with functions -3 marks)

15.Explain the plastic casting process. [U] / [CO1, CO4]
Plastic casting is a manufacturing process in which a liquid plastic material is poured into a mould and allowed to solidify into the desired shape. The process starts with preparing a suitable mould made of materials such as metal, silicone, or plaster. A liquid resin, usually a thermosetting polymer like epoxy, polyester, or polyurethane, is then mixed with a curing agent, along with pigments or fillers if required. This mixture is carefully poured into the mould

while avoiding air bubbles, sometimes using vacuum or pressure techniques to improve quality. The mould is then left undisturbed for curing, during which a chemical reaction converts the liquid into a solid plastic; this may occur at room temperature or require heating. After complete hardening, the изделие is removed from the mould and finished by trimming or polishing if necessary, making plastic casting a simple and effective method for producing complex shapes with good surface finish. (Any three points – 3 marks)

16.What is anthracene, and where is it applied? [U] / [CO5]
Anthracene is a solid polycyclic aromatic hydrocarbon composed of three fused benzene rings, typically obtained from coal tar. It appears as colorless crystals and is known for its fluorescence under ultraviolet light. Anthracene is widely used in the manufacture of dyes, particularly for producing anthraquinone-based dyes, and in the synthesis of organic chemicals. It is also used in scintillation counters for detecting radiation, in organic semiconductors, and in research related to photochemistry and material science.
(Structure/Explanation -1.5 marks,Uses – 1.5 marks)

Part C(Short Essay Questions)

Answer any 4 questions.Each question carries 7 marks

17.Analyze the role of carbon fibers in enhancing the mechanical properties of composite materials. [An] / [CO1]
Carbon nanofibers are nanoscale carbon-based fibers, typically with diameters ranging from tens to hundreds of nano meters and lengths in the micro meter range, characterized by a high aspect ratio, graphitic structure, and exceptional mechanical, thermal, and electrical properties. Carbon fibers play a crucial role in enhancing the mechanical properties of composite materials due to their high tensile strength, excellent stiffness, low density, and superior load-bearing capability. When incorporated into a matrix such as epoxy, they act as the primary reinforcement phase, carrying most of the applied stress and thereby significantly improving the strength and rigidity of the composite while keeping it lightweight. These composites also exhibit excellent fatigue and creep resistance, enabling them to withstand repeated and long-term loading without significant degradation. Their anisotropic nature allows engineers to tailor mechanical properties by aligning fibers in specific directions according to load requirements, resulting in optimized structural performance. In addition to conventional carbon fibers, carbon nanofibers have gained importance as advanced reinforcement materials due to their high surface area and strong interfacial interaction with the matrix, which enhances stress transfer and improves properties such as toughness, modulus, and crack resistance at the nanoscale. Furthermore, both carbon fibers and carbon nanofibers contribute to thermal stability and corrosion resistance, ensuring durability in harsh environments. However, despite these advantages, carbon-based composites can be relatively expensive and may exhibit brittleness and lower impact resistance. The incorporation of carbon fibers and carbon nanofibers leads to high-performance composites with superior mechanical properties suitable for advanced engineering applications.
(Carbon nanofibers definition – 2 marks, Properties with explanation – 5 marks)
18.Explain the properties and manufacturing process of ceramics. [U] / [CO2]
Ceramics are inorganic, non-metallic materials typically composed of oxides, carbides, nitrides, and silicates, formed by shaping and then hardening at high temperatures. They are widely used in engineering and structural applications due to their unique properties.
Properties: Ceramics exhibit high hardness and compressive strength, making them highly resistant to wear and deformation. They possess excellent thermal stability and can withstand

very high temperatures, which makes them suitable for refractory applications. They generally have low electrical and thermal conductivity, making them good insulators. Ceramics are also highly resistant to corrosion and chemical attack. However, they are brittle, with low tensile strength and poor impact resistance, often leading to sudden fracture.
Manufacturing process:

1. Raw material preparation**:** Selection and purification of ceramic powders, followed by mixing with additives such as binders and plasticizers to improve handling and shaping.
2. Forming (shaping): The prepared material is shaped into the desired form using methods like pressing, slip casting, extrusion, or injection molding.
3. Drying: Removal of moisture from the shaped body to prevent cracks or defects during heating.
4. Firing (sintering): Heating the dried material at high temperatures in a kiln to densify and strengthen it through particle bonding.
5. Finishing: Final operations such as grinding, glazing, or machining to achieve required surface finish, dimensions, and properties.
   (Properties – 2 marks, Manufacture – 5 marks)

19.Describe the working of a lithium ion battery and its uses. [K] / [CO3]
Working:A Lithium-ion battery works based on the reversible movement of lithium ions between two electrodes through an electrolyte. It consists mainly of a graphite anode, a lithium metal oxide cathode (such as $LiCoO_2$), an electrolyte (typically a lithium salt in an organic solvent), and a separator. During discharge (when the battery is in use), lithium atoms in the anode undergo oxidation and release electrons and lithium ions. The electrons travel through the external circuit, providing electrical energy, while the lithium ions move through the electrolyte to the cathode. At the cathode, these ions combine with electrons and undergo reduction. During charging, an external power source forces the reverse process: lithium ions move back from the cathode to the anode, and energy is stored in the battery. This continuous intercalation and deintercalation of lithium ions between the electrodes enables repeated charge–discharge cycles with high efficiency.(4 marks)
Uses: Lithium-ion batteries are widely used in modern technology due to their high energy density, lightweight nature, and long cycle life. They are commonly found in portable electronic devices such as smartphones, laptops, and tablets. They are also extensively used in electric vehicles, renewable energy storage systems, power tools, and medical devices. Their ability to provide reliable and rechargeable energy makes them essential in both consumer electronics and large-scale energy applications.(3 marks)

20.Explain the measures used to describe the performance of a battery.[U] / [CO3]
The performance of a battery can be described using the following measures
1.Capacity: The total amount of charge a battery can store and deliver, usually expressed in Ah or mAh.
2.Energy density: The amount of energy stored per unit mass or volume (Wh/kg or Wh/L), indicating how lightweight and compact the battery is.
3.Power density: The rate at which energy can be delivered, important for high-power applications.
4.Voltage: The potential difference between electrodes that drives the flow of electrons.
5.Efficiency: Indicates how effectively the battery operates; includes coulombic efficiency and energy efficiency.
6.Cycle life: The number of charge–discharge cycles the battery can undergo before significant capacity loss.
7.Self-discharge rate: The rate at which the battery loses charge when not in use.

8.Internal resistance: The opposition to the flow of current within the battery, affecting heat generation and performance.
9.State of Charge (SOC): The current level of charge in the battery relative to its capacity.
10.State of Health (SOH): The overall condition and aging status of the battery compared to a new one.
(Any five points with description – 7marks)

21.Explain the process of polymerization in the formation of thermoplastic and thermosetting polymers.[K] / [CO4]
Polymerization is the chemical process by which small molecules called monomers combine to form large macromolecules (polymers), and the nature of this process determines whether a material becomes a thermoplastic or a thermosetting polymer. In thermoplastic polymers, polymerization typically occurs through addition (chain-growth) or condensation (step-growth) reactions that produce mostly linear or slightly branched chains without extensive cross-linking. In addition polymerization, unsaturated monomers (like ethylene) react by opening their double bonds and linking together in a chain reaction, while in condensation polymerization, monomers combine with the elimination of small molecules such as water or methanol. Because these polymers have little or no cross-linking, the intermolecular forces between chains are relatively weak, allowing the material to soften upon heating and harden upon cooling repeatedly without significant chemical change.
In thermosetting polymers are formed through polymerization processes that create a highly cross-linked, three-dimensional network structure. Initially, monomers or low molecular weight prepolymers react (often through condensation or addition mechanisms) to form a viscous or soft intermediate, which upon further heating or curing undergoes extensive cross-linking. This curing process forms strong covalent bonds between polymer chains, resulting in a rigid, infusible, and insoluble structure. Once set, thermosetting polymers cannot be remelted or reshaped because heating does not soften them but instead leads to decomposition. Thus, the key difference lies in the degree of cross-linking formed during polymerization, which governs the thermal and mechanical behavior of thermoplastics and thermosets. (Explanation of both polymers – 7 marks)
22.Describe the ultimate analysis of coal and how it differs from proximate analysis.[U] / [CO5]
Ultimate analysis of coal is a detailed chemical analysis that determines the elemental composition of coal, namely the percentages of carbon (C), hydrogen (H), nitrogen (N), sulfur (S), and oxygen (O). Carbon and hydrogen are usually estimated by burning a known mass of coal in oxygen and measuring the amounts of carbon dioxide and water formed. Nitrogen is commonly determined by methods such as the Kjeldahl method, sulfur is estimated by converting it into sulfate and measuring it, and oxygen is calculated by difference. This analysis provides precise information about the chemical makeup of coal and is useful for calculating calorific value, combustion requirements, and assessing environmental impact.
In contrast, proximate analysis is a simpler and quicker method that determines the physical constituents of coal, namely moisture content, volatile matter, ash content, and fixed carbon. It involves controlled heating of coal under specified conditions to measure weight loss and residue. While proximate analysis gives a general idea about the fuel behavior during combustion, ultimate analysis gives a more detailed and quantitative understanding of the elemental composition. Thus, the main difference is that ultimate analysis deals with chemical composition, whereas proximate analysis deals with physical components and combustion characteristics.
(Ultimate analysis – 3.5 marks, Proximate analysis – 3.5 marks)